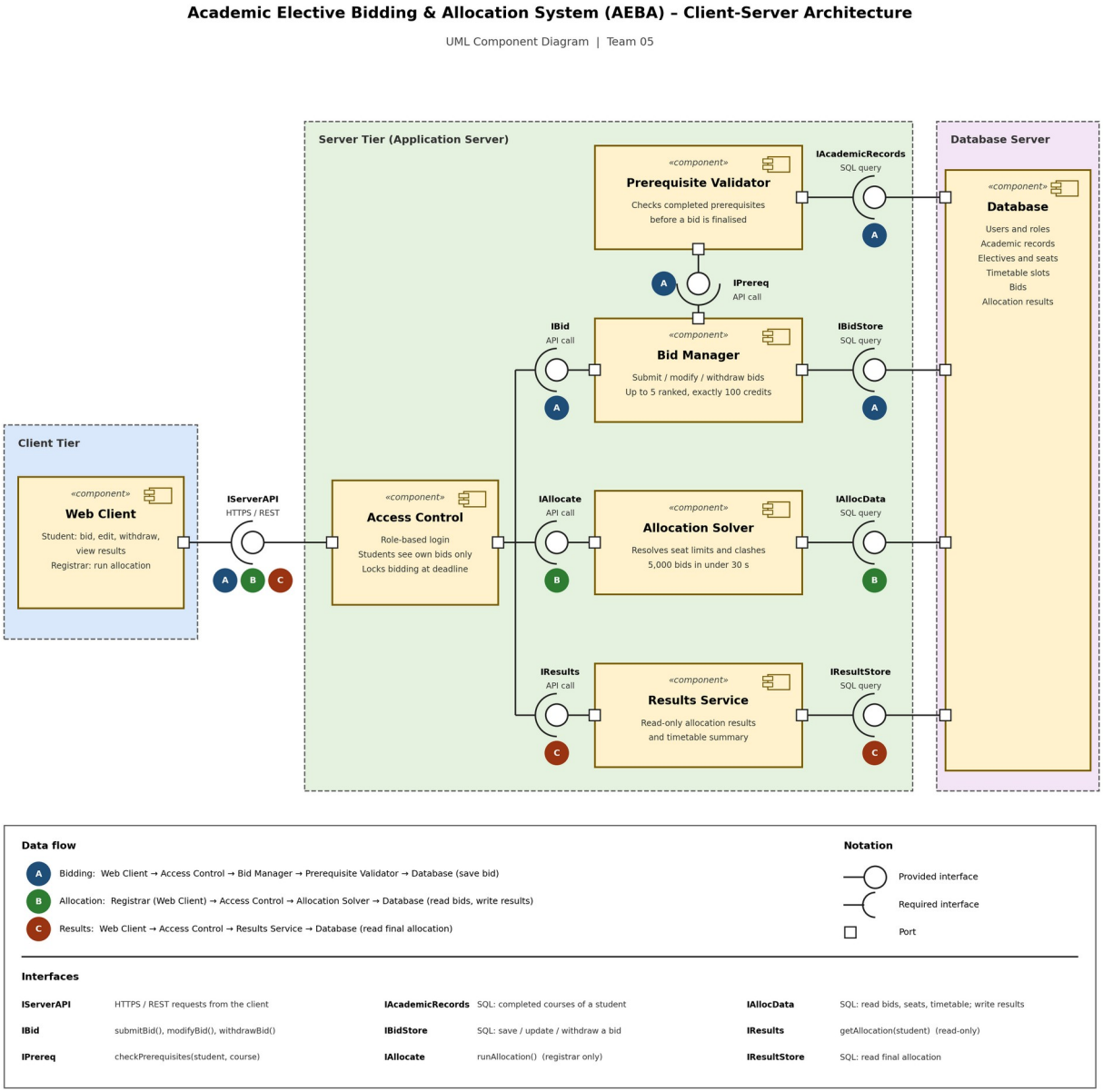
Lab 3: Component Modelling & Architectural Pattern

Team 05: Academic Elective Bidding and Allocation System

PES1UG24AM156



Architecture Selection and Justification

System: Academic Elective Bidding & Allocation System (AEBA)
Selected Architectural Pattern: Client-Server Architecture

1. Architectural Choice

We chose **Client-Server Architecture** for the Academic Elective Bidding & Allocation System. Students and the registrar use the Web Client in their browser. The business logic runs on one central application server (Access Control, Bid Manager, Prerequisite Validator, Allocation Solver and Results Service) and the data is stored in the Database on a separate database server. The client never talks to the database directly. Every request goes to Access Control through the IServerAPI interface first and is then passed to the right component.

We also looked at Layered and Microservices. Layered would work, but it does not say where the heavy allocation work runs, and Microservices would spread bids and results over many services, which makes it harder to keep the data consistent for a system of this size.

2. Reasons for Selection

Reason 1 — Data stays in one place: FR-001 needs exactly 100 credits spread over up to 5 ranked courses, and FR-003 says seats can never be overbooked. With one server and one database these checks happen in a single place, so every student is treated the same way and there are no copies of data to keep in sync.

Reason 2 — One server decides the rules and the deadline: FR-005 and NFR-002 say a bid cannot be changed after the deadline. If the server uses its own clock to lock bidding, a student cannot get around it by editing the page in the browser. The same server also runs the prerequisite check (FR-002) whenever a bid is submitted or changed, and it is simple to deploy because there is only one server to host and update.

3. Security Advantage

All access goes through the Access Control component, so each request is checked for the user's role before anything else happens (NFR-002). A student can only see and change their own bids, only the registrar can start the allocation, and the database is never exposed to the browser.

4. Performance Benefit

The Allocation Solver runs on the server close to the database, so it reads all the bids, seat limits and timetable data locally instead of sending them over the network. This helps with NFR-001 (5,000 bids in under 30 seconds), and the server can be given more CPU and memory for the allocation run at the end of the bidding window.

Conclusion

Client-Server fits AEBA because it keeps the bidding data and rules in one controlled place and keeps the heavy allocation work on the server. The main downside is that the server is a single point of failure, which we would reduce with regular database backups. The allocation only runs once after the deadline, so the load is easy to predict.